

LUCA VIALETTTO

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 Director, Multiscale Computational Plasma Laboratory
 University of California, Los Angeles

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RESEARCH VISION AND INTERESTS

I develop multiscale computational frameworks for nonequilibrium plasmas and plasma–material interactions. My research connects kinetic plasma physics, transport, surface science, and data-driven methods across scales, from microscopic particle–surface interactions to device and reactor behavior. I aim to develop predictive, physics-based models that replace empirical boundary conditions and trial-and-error approaches with quantitative descriptions of complex plasma–material interfaces. My methods integrate particle, Boltzmann, and fluid simulations with surface kinetics, data assimilation, machine learning, and high-performance computing. Applications include electric propulsion and advanced plasma processing.

EDUCATION

Nov 2021	Ph.D. in Applied Physics , <i>summa cum laude</i> (top 5%) Eindhoven University of Technology, Netherlands
Sep 2017	M.Sc. in Physics , <i>summa cum laude</i> University of Padova, Italy
Sep 2015	B.Sc. in Physics University of Padova, Italy

EMPLOYMENT HISTORY

Jul 2026–present	Assistant Professor Mechanical and Aerospace Engineering, University of California, Los Angeles
Sep 2023–Jun 2026	Stanford Energy Postdoctoral Fellow Plasma Dynamics Modeling Laboratory Aeronautics and Astronautics, Stanford University
Jan 2022–Jul 2023	Postdoctoral Researcher Theoretical Electrical Engineering, Kiel University, Germany
Oct 2017–Nov 2021	Ph.D. Researcher Computational Plasma Physics and Chemistry Group Dutch Institute for Fundamental Energy Research, Eindhoven, Netherlands

Mar 2017–Sep 2017	M.Sc. Research Fellow Plasma for Gas Conversion Group Max Planck Institute for Plasma Physics, Garching, Germany
Mar 2015–Sep 2015	B.Sc. Research Fellow Consorzio RFX, National Research Council (CNR), Padova, Italy

AWARDS AND HONORS

2025	Outstanding Reviewer Award , Plasma Sources Science and Technology Journal
2023	Rutherford Plasma Physics Communication Prize , Institute of Physics (IOP)
2022	Trusted Reviewer , IOP Publishing
2021	Doctoral Program Thesis Distinction (top 5%), Eindhoven University of Technology
2021	Student Excellence Award , 74th Gaseous Electronics Conference (GEC 2021), American Physical Society
2018	Poster Prize for Plasma Modelling , Europhysics Conference on the Atomic and Molecular Physics of Ionized Gases (ESCAMPIG), European Physical Society

RESEARCH SUPPORT AND FELLOWSHIPS

2026–2027	Princeton Collaborative Low Temperature Plasma Research Facility (PCRF) User Facility Award , U.S. Department of Energy, Princeton Plasma Physics Laboratory (PPPL) Competitive proposal awarded experimental facility access for validation of a data-driven collisional-radiative model for RF-driven plasmas; with K. Hara (Stanford University), technical point of contact Y. Raitses (PPPL).
2023–2026	Stanford Energy Postdoctoral Fellowship , Stanford University (approx. \$280,000) Inaugural cohort; competitive fellowship supporting independent research on plasma-assisted nitrogen fixation and plasma–liquid interactions.
2022	Grant from the European Integrated Research Training Center (approx. \$10,000) Research visit to the laboratory of Prof. S. Hamaguchi, Osaka University, Japan
2017	Scholarship from the Erasmus Exchange European Program Research stay at the Max Planck Institute for Plasma Physics

TEACHING EXPERIENCE

Instructor

Fall 2026	Transport Phenomena, University of California, Los Angeles
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Teaching assistant

Winter 2022	Modeling and Simulations of Complex Physical Systems, Kiel University
2018–2020	Gas Discharges (3MP100), Eindhoven University of Technology

RESEARCH SUPERVISION

Ph.D. students

Jinghua Zhang	Ph.D. 2026–, Mechanical and Aerospace Engineering, UCLA
Natalia F. de La Torre	Ph.D. 2026–, Mechanical and Aerospace Engineering, UCLA
Elena Ancona	Ph.D. 2023–2025, Politecnico di Bari, Italy (co-supervisor)

Master’s students

Christian Stüwe	M.S. 2022–2023, Kiel University, Germany
Adel Ben Moussa	M.S. 2020, Mines ParisTech, France

Undergraduate researchers

Ryan Chick	2026–, Mechanical and Aerospace Engineering, UCLA
Camilly P. dos Santos	SUPER 2026, Aeronautics and Astronautics, Stanford University
Daisy Li	SUPER 2025, Aeronautics and Astronautics, Stanford University
Zach Trace Green	SUPER 2024, Aeronautics and Astronautics, Stanford University
Patrick Dietrich	B.Sc. 2023, Kiel University, Germany
Luciano Baldassi	B.Sc. 2020–2021, Eindhoven University of Technology, Netherlands

Thesis committees

Gonçalo Amadeu	M.Sc.; Advisor: V. Guerra, Instituto Superior Técnico, Lisbon; 2024 (with M. J. Kushner)
Mendes Cardoso	Kushner)
Jorge Nuno Machado	M.Sc.; Advisor: V. Guerra, Instituto Superior Técnico, Lisbon; 2023 (with A. S. Morillo-Candás)
Silveira	

RESEARCH SOFTWARE AND COMPUTATIONAL METHODS

Research software: Development of scalable computational frameworks for kinetic and continuum plasma simulation, particle transport, surface kinetics, and data-driven plasma modeling using C++, Python/JAX, MPI, PETSc, and high-performance computing.

PUBLICATIONS AND PRESENTATIONS

* corresponding author; † equal contribution.

Journal Articles

27. T. Gergs*, R. Lamprecht, S. Yarragolla, O. Gronenberg, **L. Vialetto**, H. Kohlstedt, T. Mussenbrock, and J. Trieschmann*, “From Processing to Functionality: Engineering Accessible Material States in Cu-Embedded SiO_x Memristive Devices”, *submitted* [DOI]
26. T. Lafleur*, B. Esteves, C. Drag, A. Bourdon, P. Chabert, J. Martínez Martínez, **L. Vialetto**, G. Bowden, and R. Shaik, “Iodine plasmas for space propulsion and industrial applications”, *Journal of Physics D: Applied Physics* **59**, 023001 (2026) [DOI]

25. **L. Vialletto*** and K. Hara, “Non-stationary multi-term approach for the electron Boltzmann equation in pulsed plasmas”, *Physics of Plasmas* **33**, 043501 (2026) [\[DOI\]](#)
24. **L. Vialletto*** and K. Hara, “Three-body electron attachment in humid air”, *Plasma Sources Science and Technology* **34**, 035006 (2025) [\[DOI\]](#)
23. **L. Vialletto***, E. Ancona, and S. Longo, “A variance reduction technique for Monte Carlo simulations of electrons and ions in electric and magnetic fields”, *Physics of Plasmas* **32**, 013903 (2025) [\[DOI\]](#)
22. R. Lamprecht, **L. Vialletto**, T. Gergs, F. Zahari, R. Marquardt, H. Kohlstedt, and J. Trieschmann*, “Thickness-related analog switching in SiO_x/Cu/SiO_x memristive devices for neuromorphic applications”, *Advanced Engineering Materials* **27**, 2401824 (2025) [\[DOI\]](#)
21. A. R. Mansour*, **L. Vialletto**, Y. Yamashita, and K. Hara, “Electron inertial effects in the rarefied regime of a direct-current (DC) breakdown”, *Plasma Sources Science and Technology* **33**, 115018 (2024) [\[DOI\]](#)
20. M. Altin, P. Viegas, **L. Vialletto**, G. J. van Rooij, and P. Diomede*, “Spatio-temporal analysis of power deposition and vibrational excitation in pulsed N₂ microwave discharges from 1D fluid modelling and experiments”, *Plasma Sources Science and Technology* **33**, 045008 (2024) [\[DOI\]](#)
19. M. Flynn*, **L. Vialletto**, A. Fierro, A. Neuber, and J. Stephens, “Benchmark calculations for anisotropic scattering in kinetic models for low temperature plasma”, *Journal of Physics D: Applied Physics* **57**, 255204 (2024) [\[DOI\]](#)
18. **L. Vialletto***, H. Sugawara, and S. Longo, “Particle propagation and electron transport in gases”, *Plasma* **7**, 121-145 (2024) [\[DOI\]](#)
17. J. Afonso, **L. Vialletto**, V. Guerra, and P. Viegas*, “Plasma-induced reversible surface modification and its impact on oxygen heterogeneous recombination”, *Journal of Physics D: Applied Physics* **57**, 04LT01 (2024) [\[DOI\]](#)
16. J. Trieschmann*, **L. Vialletto**, and T. Gergs, “Review: Machine learning for advancing low-temperature plasma modeling and simulation”, *Journal of Micro/Nanopatterning, Materials, and Metrology* **22**, 041504 (2023) [\[DOI\]](#)
15. M. Budde*, T. Dias, **L. Vialletto**, N. Pinhão, V. Guerra, and T. Silva, “Electron-neutral collision cross sections for H₂O: II. Anisotropic scattering and assessment of the validity of the two-term approximation”, *Journal of Physics D: Applied Physics* **56**, 255201 (2023) [\[DOI\]](#)
14. M. Altin, **L. Vialletto**, S. Longo, P. Viegas, and P. Diomede*, “A modified Fokker-Planck approach for a complete description of vibrational kinetics in a N₂ plasma chemistry model”, *The Journal of Physical Chemistry A* **127**, 261-275 (2023) [\[DOI\]](#)
13. M. Altin, P. Viegas, **L. Vialletto**, A. W. van de Steeg, S. Longo, G. J. van Rooij, and P. Diomede*, “Energy partitioning in N₂ microwave discharges: integrated Fokker-Planck approach to vibrational kinetics and comparison with experiments”, *Plasma Sources Science and Technology* **31**, 104003 (2022) [\[DOI\]](#)
12. M. Budde*, T. Dias, **L. Vialletto**, N. Pinhão, V. Guerra, and T. Silva, “Electron-neutral collision cross sections for H₂O: I. Complete and consistent set”, *Journal of Physics D: Applied Physics* **55**, 445205 (2022) [\[DOI\]](#)
11. **L. Vialletto***, A. W. van de Steeg, P. Viegas, S. Longo, G. J. van Rooij, M. C. M. van de Sanden, J. van Dijk, and P. Diomede, “Charged particle kinetics and gas heating in CO₂ microwave plasma contraction:

- comparisons of simulations and experiments”, *Plasma Sources Science and Technology* **31**, 055005 (2022) [\[DOI\]](#)
10. V. Laporta*, **L. Vialetto**, and V. Guerra, “Vibrational excitation cross sections for non-equilibrium nitric oxide-containing plasma”, *Plasma Sources Science and Technology* **31**, 054001 (2022) [\[DOI\]](#)
 9. A. W. van de Steeg†, **L. Vialetto**†, A. F. Silva, P. Viegas, P. Diomede, M. C. M. van de Sanden, and G. J. van Rooij*, “The chemical origins of plasma contraction and thermalization in CO₂ microwave discharges”, *The Journal of Physical Chemistry Letters* **13**, 1203-1208 (2022) (†equal contribution) [\[DOI\]](#)
 8. **L. Vialetto***, A. Ben Moussa, J. van Dijk, S. Longo, P. Diomede, V. Guerra, and L. L. Alves, “Effect of anisotropic scattering for rotational collisions on electron transport parameters in CO”, *Plasma Sources Science and Technology* **30**, 075001 (2021) [\[DOI\]](#)
 7. P. Viegas*, **L. Vialetto**, A. W. van de Steeg, A. J. Wolf, W. A. Bongers, G. J. van Rooij, M. C. M. van de Sanden, P. Diomede, and F. J. J. Peeters, “Resolving discharge parameters from atomic oxygen emission”, *Plasma Sources Science and Technology* **30**, 065022 (2021) [\[DOI\]](#)
 6. G. Micca Longo, **L. Vialetto**, P. Diomede, S. Longo*, and V. Laporta, “Plasma modelling and prebiotic chemistry: a review of the state of the art and perspective”, *Molecules* **26**, 3663 (2021) [\[DOI\]](#)
 5. A. W. van de Steeg, **L. Vialetto**, A. F. Silva, F. J. J. Peeters, D. C. M. van den Bekerom, N. Gatti, P. Diomede, M. C. M. van de Sanden, and G. J. van Rooij*, “Revisiting spontaneous Raman scattering for direct oxygen atom quantification”, *Optics Letters* **46**, 2172-2175 (2021) [\[DOI\]](#)
 4. **L. Vialetto***, P. Viegas, S. Longo, and P. Diomede, “Benchmarking of Monte Carlo flux simulations of electrons in CO₂”, *Plasma Sources Science and Technology* **29**, 115006 (2020) [\[DOI\]](#)
 3. P. Viegas*, **L. Vialetto**, A. J. Wolf, F. J. J. Peeters, P. W. C. Groen, T. W. H. Righart, W. A. Bongers, M. C. M. van de Sanden, and P. Diomede, “Insight into contraction dynamics of microwave plasmas for CO₂ conversion from plasma chemistry modelling”, *Plasma Sources Science and Technology* **29**, 105014 (2020) [\[DOI\]](#)
 2. **L. Vialetto**, S. Longo, and P. Diomede*, “Benchmark calculations for electron velocity distribution function obtained with Monte Carlo flux simulations”, *Plasma Sources Science and Technology* **28**, 115015 (2019) [\[DOI\]](#)
 1. M. Barbisan*, C. Baltador, B. Zaniol, M. Cavenago, U. Fantz, R. Pasqualotto, G. Serianni, **L. Vialetto**, and D. Wunderlich, “First hydrogen operation of NIO1: characterization of the source plasma by means of an optical emission spectroscopy diagnostic”, *Review of Scientific Instruments* **87**, 02B319 (2016) [\[DOI\]](#)

Manuscripts in Preparation

4. **L. Vialetto**, J. de Urquijo, O. González-Magaña, B. Esteves, and A. Bourdon, “Electron swarm analysis for molecular iodine: comparison of experiments and simulations” (in preparation)
3. **L. Vialetto**, P. Barnwal, A. Dwivedi, R. M. Sankaran, and K. Hara, “Data assimilation of optical emission spectroscopy and collisional-radiative model for an atmospheric-pressure plasma jet” (in preparation)
2. **L. Vialetto**, Y. Ussenov, M. Da Valle, M. Morasco, S. Thompson, A. Yalin, Y. Raitses, and K. Hara, “Data assimilation of optical emission spectroscopy and collisional-radiative model for an inductively coupled plasma source” (in preparation)

1. **L. Vialetto**, “A framework for multiscale simulations of plasma-surface interactions for air-breathing electric propulsion”, AIAA SciTech Forum 2027, Orlando, FL, January 2027 (in preparation)

Invited Talks and Seminars

15. **L. Vialetto**, P. Barnwal, A. Dwivedi, R. M. Sankaran, and K. Hara, “Data assimilation for plasma and surface kinetics”, 53rd IEEE International Conference on Plasma Science (ICOPS 2026), Lake Tahoe, NV, June 2026 [\[DOI\]](#)
14. **L. Vialetto**, “Plasma chemistry for space propulsion and industrial applications”, AA559 Plasma Science Seminar, University of Washington, Seattle, WA, February 2026
13. **L. Vialetto** and K. Hara, “Electron kinetics in swarms and plasmas”, 78th Gaseous Electronics Conference (GEC 2025), Seoul, Republic of Korea, October 2025
12. **L. Vialetto**, “Volume and surface kinetics of plasmas for semiconductor fabrication”, MagNetUS Workshop, Morgantown, WV, June 2025
11. **L. Vialetto** and K. Hara, “Plasma chemistry for energy and sustainability”, International Conference on Fundamental Plasma Physics (ICFPP), Austin, TX, March 2025
10. **L. Vialetto**, “The new era of electric propulsion: plasma meets advanced computing”, Department of Mechanical and Aerospace Engineering, University of California, Los Angeles, CA, February 2025
9. **L. Vialetto**, “Particle propagation and electron transport in gases”, 32nd Summer School and International Symposium on the Physics of Ionized Gases (SPIG 2024), Belgrade, Serbia, August 2024
8. **L. Vialetto**, “Challenges in plasma modelling for environmental applications: electron kinetics and transport”, XXVI Europhysics Conference on the Atomic and Molecular Physics of Ionized Gases (ESCAMPIG 2024), Brno, Czech Republic, July 2024
7. **L. Vialetto**, “Modeling plasmas for CO₂ conversion: from fundamental data to applications”, International Online Plasma Seminar (IOPS) and Online Low Temperature Plasma (OLTP) Seminar, remote, December 2023
6. **L. Vialetto**, C. Stüwe, T. Gergs, T. Mussenbrock, and J. Trieschmann, “Multiscale modeling of plasma-surface interactions”, 49th European Physical Society Conference on Plasma Physics (EPS 2023), Bordeaux, France, July 2023
5. **L. Vialetto**, “Modeling plasmas for environmental applications”, Laboratoire de Physique des Plasmas (LPP), École Polytechnique, Palaiseau, France, April 2023
4. **L. Vialetto**, P. Viegas, M. Altin, S. Longo, J. van Dijk, and P. Diomede, “Modeling electron kinetics, chemistry, and transport in microwave plasmas for conversion of CO₂ and N₂”, MGK Colloquium, Ulm, Germany, March 2023
3. **L. Vialetto** and A. Ho, “Rutherford Plasma Physics Communication Prize: CoffeeBreakDown Podcast”, 49th IOP Annual Plasma Physics Conference, Oxford, UK, remote, March 2023
2. **L. Vialetto** and L. L. Alves, “Anisotropic scattering in rotational collisions of electrons with CO molecules”, XXII International Symposium on Electron-Molecule Collisions and Swarms (POSMOL 2021), remote, July 2021

1. **L. Vialetto**, P. Viegas, S. Longo, and P. Diomede, “Monte Carlo flux modelling of electron kinetics in CO₂”, 30th International Symposium on the Physics of Ionized Gases (SPIG 2020), remote, August 2020

Conference Presentations

12. **L. Vialetto** and K. Hara, “Non-stationary multi-term Boltzmann solver for electron kinetics in pulsed plasmas”, 53rd IEEE International Conference on Plasma Science (ICOPS 2026), Lake Tahoe, NV, June 2026 [\[DOI\]](#)
11. **L. Vialetto** and K. Hara, “Towards integrated modeling of the plasma-material interface”, 67th Annual Meeting of the APS Division of Plasma Physics, Long Beach, CA, November 2025 (poster)
10. **L. Vialetto** and K. Hara, “Fluid modeling of surface dielectric barrier discharges to understand bulk and surface phenomena”, 77th Gaseous Electronics Conference (GEC 2024), San Diego, CA, FR1.00001, October 2024
9. **L. Vialetto**, C. Stüwe, T. Gergs, T. Mussenbrock, and J. Trieschmann, “Multiscale modeling of plasma-surface interactions”, 35th International Conference on Phenomena in Ionized Gases (ICPIG XXXV), Egmond aan Zee, Netherlands, July 2023
8. **L. Vialetto**, R. Lamprecht, C. Stüwe, T. Hemke, F. Zahari, H. Kohlstedt, T. Mussenbrock, and J. Trieschmann, “Multiscale transport modeling of reactive sputtering for fabrication of neuromorphic hardware”, 75th Gaseous Electronics Conference (GEC 2022), Sendai, Japan, DF1.00002, October 2022
7. **L. Vialetto**, A. W. van de Steeg, G. J. van Rooij, P. Diomede, and P. Viegas, “Modelling chemistry and transport in microwave plasma discharges for CO₂ conversion”, XXV Europhysics Conference on the Atomic and Molecular Physics of Ionized Gases (ESCAMPIG 2022), Paris, France, July 2022
6. **L. Vialetto**, P. Viegas, A. W. van de Steeg, G. J. van Rooij, J. van Dijk, S. Longo, and P. Diomede, “Student Excellence Award Finalist: Insight into chemistry and transport in CO₂ microwave discharges through comparisons between simulations and experiments”, 74th Gaseous Electronics Conference (GEC 2021), virtual, DT24.00008, October 2021 (Awarded)
5. **L. Vialetto**, P. Viegas, S. Longo, and P. Diomede, “Monte Carlo flux simulations of electrons in CO₂”, Physics@Veldhoven, virtual, PT6B.3, January 2021
4. **L. Vialetto**, P. Viegas, S. Longo, and P. Diomede, “Fast and accurate simulations of electrons in CO₂ using Monte Carlo flux”, 73rd Gaseous Electronics Conference (GEC 2020), virtual, MW3.00004, October 2020
3. **L. Vialetto**, S. Longo, and P. Diomede, “Monte Carlo flux simulations of electrons for plasma modelling”, 21st Workshop on the Exploration of Low-Temperature Plasma Physics (WELTPP-21), Kerkrade, Netherlands, O13, November 2019
2. **L. Vialetto**, S. Longo, and P. Diomede, “Monte Carlo flux simulations of electrons for plasma modelling”, 34th International Conference on Phenomena in Ionized Gases (ICPIG) and 10th International Conference on Reactive Plasmas (ICRP), Sapporo, Japan, OR16PM-D01, July 2019
1. **L. Vialetto**, S. Longo, and P. Diomede, “Monte Carlo flux simulations of electrons for plasma modelling”, 31st Symposium on Plasma Physics and Radiation Technology, Lunteren, Netherlands, O2, March 2019

Ph.D. Dissertation

- “Modelling of Plasma for CO₂ Conversion: Electron Kinetics, Chemistry and Transport.” Advisors: M. C. M. (Richard) van de Sanden and P. Diomede; Eindhoven University of Technology, 2021 (*summa cum laude*, top 5%)

PROFESSIONAL AFFILIATIONS

- Affiliated Faculty (2026–), Semiconductor Hub, UCLA Samueli School of Engineering
- Member (2024–), IEEE Nuclear and Plasma Sciences Society
- Member, APS, Gaseous Electronics Conference
- Member, EPS Plasma Physics Division

PROFESSIONAL SERVICE

Societies

- Vice-Chair, IEEE Nuclear and Plasma Sciences Society Young Professionals (2026–)
- Member, LXCat team (2024–), international open-access database and community resource for low-temperature plasma modeling and experimental data

Conferences

- Session chair, “Plasma Modeling”, 79th Gaseous Electronics Conference (GEC 2026), Chicago, IL (upcoming, November 2026)
- Session chair, “Plasma Chemistry”, 53rd IEEE International Conference on Plasma Science (ICOPS 2026), Lake Tahoe, NV (2026)
- Session organizer, “Basic Processes in Fully/Partially Ionized Plasmas”, 50th IEEE International Conference on Plasma Science (ICOPS 2023), Santa Fe, NM (2023)
- Session chair, “Chemical Synthesis by Plasma”, 77th Gaseous Electronics Conference (GEC 2024), San Diego, CA (2024)
- Session chair, “Plasma for Environmental Applications”, 35th International Conference on Phenomena in Ionized Gases (ICPIG XXXV), Egmond aan Zee, Netherlands (2023)
- Organizer, educational workshop “All about the electron Boltzmann equation”, Dutch Institute for Fundamental Energy Research, Netherlands (2019); organizing committee: L. Vialetto, V. Guerra, P. Diomede, and P. Viegas

Peer Review

- Journal: Plasma Sources Science and Technology (IOP Trusted Reviewer 2022); Journal of Physics D: Applied Physics; Physics of Plasmas; Scientific Reports; Plasma Chemistry and Plasma Processing; Journal of CO₂ Utilization; Machine Learning: Science and Technology. 45 verified peer reviews as of September 2026